

Unsupervised Learning - Lab 07 - DBSCAN Clustering

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Agenda

- Motivation:
 - This seventh lab session aims to design and apply the Density-Based Spatial Clustering of Applications with Noise (DBSCAN) algorithm to an unknown dataset.
- Steps:
 1. You will not be given the parameter K_d , then you need to guess an appropriate number of clusters by analysis.
 2. Guess the number of clusters K , if any, in the dataset from visual inspection.
 3. Apply DBSCAN in a naive way and check results (find the number of K_d clusters formed by the algorithm and visualize the clustering solution).
 4. After identifying the "knee point" using the "knee method", apply DBSCAN again and compare the new results (any change in the number of K_d clusters formed?).
 5. Validate the clustering solution obtained at point 3, using one or more measures/strategies to validate a clustering solution (e.g., inter-cluster distances, silhouette score, etc.).
 6. Use the true labels to further validate the DBSCAN solution using supervised measures.

- You are going to re-use already developed code: check solutions of previous labs.
- Also, you are going to add code to properly select the main parameters of DBSCAN (ϵ , i.e., the radius, and *MinPts*, i.e., the minimum number of points required to define a "dense" core region), and to test this particular algorithm.
- The use of package `sklearn.cluster.DBSCAN` is the main novelty of this lab.

- The data to work on will be available on eLearning.
- The true labels will be available on eLearning.
- Useful packages: numpy, pandas, scipy, matplotlib, seaborn, sklearn, `scipy.cluster` (NEW!)